Analysis of Overseas-Owned Property Acquisition Data and Sanctions Matching

Executive Summary

Objective	Data and tools	What this shows
Identify long-run patterns in overseas-owner acquisition events and assess whether sanctions data could be matched reliably to ownership records.	Cleaned Land Registry ownership data, cleaned sanctions data, SQL joins, regex standardisation, window functions, Jaro-Winkler similarity, and Python visualisation.	Large-scale SQL cleaning, careful entity matching, sensible validation design, and clear analytical writing without overstating weak signals.

Business Context and Objective

Essentially, the business question behind the work was straightforward: could ownership records be linked to sanctions records well enough to support further investigation into patterns of acquisition and possible post-designation ownership changes? In practice, however, that question quickly became a rigorous data-quality and matching problem. It should be noted that property ownership datasets and sanctions lists do not always share a stable universal identifier. Furthermore, name fields are frequently inconsistent, incomplete, and formatted differently across various sources.

To resolve this, I worked primarily in SQL, utilising cleaned Land Registry ownership data alongside a cleaned sanctions list and a sanctions-to-company reference table. The overall workflow combined regex-based standardisation, joins, window functions, and similarity scoring. Ultimately, once the core SQL outputs were successfully in place, Python was used to visualise the main time-series and jurisdiction patterns.

Stack: SQL and Snowflake.

Data Sources:

- UK Sanctions List
- OCOD (Overseas Companies Ownership of Data)
- Companies House PSC Data

Technical Approach

The initial phase of my methodology necessitated rigorous data preparation and validation. Before executing any matching algorithms, I methodically verified the structure and completeness of the working ownership table. In the queried sample, the cleaned ownership dataset encompassed 5,728,514 rows and, crucially, returned 0.00 per cent blank proprietor names. Consequently, this provided the required confidence that the data was robust enough for complex downstream matching and trend analysis. Furthermore, I standardised both

company registration numbers and proprietor names by trimming whitespace, normalising case, and stripping extraneous punctuation or non-alphanumeric characters. Essentially, this mitigated avoidable mismatches stemming from superficial formatting discrepancies rather than substantive differences.

Subsequently, the second stage concentrated on exact matching. Where a company number or an alternative stable identifier was present, I treated this as the optimal linkage route. I normalised the company registration numbers across both the sanctions reference data and the ownership data, deploying SQL joins to ascertain whether sanctioned entities could be definitively connected to titles within the ownership dataset. It must be noted that exact identifier matching is highly auditable; it is fundamentally more straightforward to explain, defend, and reproduce than a probabilistic name-only match. Therefore, this established an authoritative and transparent baseline for the project.

On the other hand, the third stage required fuzzy matching for instances where exact identifiers were invariably missing. I utilised Jaro-Winkler similarity scoring on the cleaned name fields. However, I did not indiscriminately compare every sanctions name against every proprietor name. Instead, I introduced strategic blocking logic by matching strictly on the first three letters of the cleaned first token and constraining the acceptable length discrepancy. While this efficiently reduced the search space and rendered the computation practical, the scale of the challenge remained apparent. Even within a constrained 500-record sanctions sample, the process still generated 758,996 candidate pairs. This vividly illustrates how rapidly fuzzy matching parameters can expand computational load, even following sensible blocking.

Ultimately, the fourth stage involved event sequencing. By leveraging SQL window functions, specifically ROW_NUMBER and LEAD, I engineered logic to identify subsequent title events occurring after the sanctions designation dates. The overarching objective was to determine whether matched entities exhibited post-designation ownership-related activity that might warrant further regulatory review. In the executed logic, these tests yielded no confirmed results. However, I treated this as a legitimate, highly valuable analytical outcome rather than a failure; it demonstrated that the constructed workflow could successfully rule out a signal under the prevailing assumptions.

Table 1. Select Validation Metrics

Rows in cleaned table **5,728,514**	Blank proprietor names **0.00%**	Fuzzy candidate pairs scored **758,996**
Candidates with score >= 85 **286**	Candidates with score >= 90 **0**	Highest fuzzy score **88**

Results and Interpretation

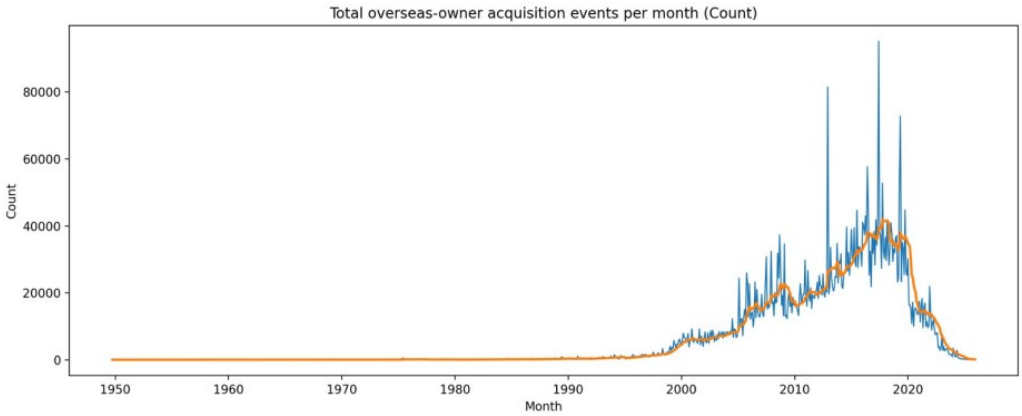
It should be noted that the validation results derived from the fuzzy matching approach were

a significant finding in their own right. Specifically, only 286 candidate pairs achieved a similarity score of at least 85, no candidate scored 90 or above, and the maximum observed score peaked at 88. Consequently, this strongly suggested that fuzzy name matching, in isolation, is insufficiently robust to support confident entity resolution within this context. Furthermore, rather than erroneously presenting weak name similarities as probable matches, I was able to demonstrate that the methodology had been rigorously tested, quantified, and appropriately treated with analytical scepticism.

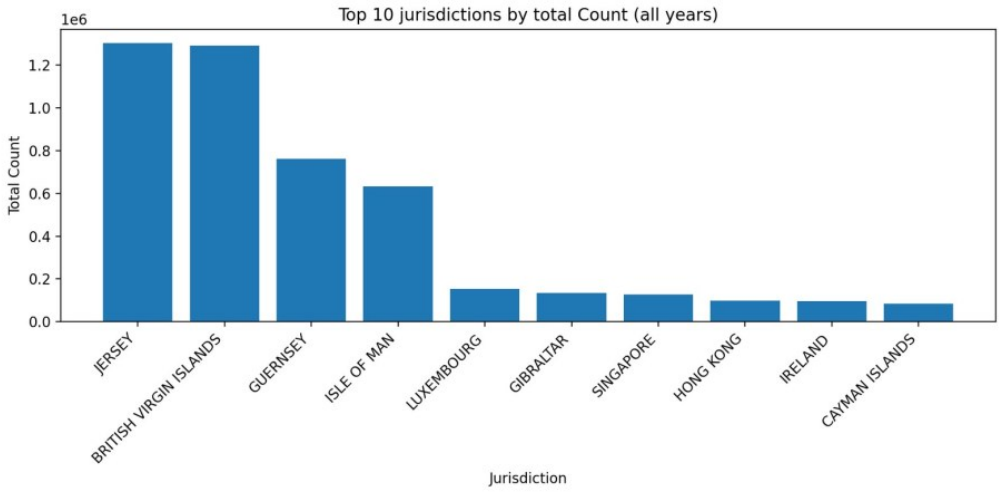
In addition to the core matching programme, I utilised the ownership data to explore long-run acquisition patterns. The resulting time-series visualisation illuminated a distinct trajectory: overseas-owner acquisition events remained ostensibly low for several decades, increased steadily from the late 1990s, accelerated throughout the 2000s, and reached a pronounced peak in the late 2010s before experiencing a sharp decline post-2020. Upon analysing jurisdiction totals, it became apparent that activity was heavily concentrated within a relatively small subset of offshore centres. Essentially, Jersey and the British Virgin Islands recorded the highest overall frequencies, closely followed by Guernsey and the Isle of Man. Please see Figures 1 and 2 for more information.

However, one of the most critical lessons extracted from this project was the absolute necessity of methodological restraint. In the realm of sanctions-related analysis, it must be noted that a weak match can often be more detrimental than no match at all, particularly if it instils false confidence. Consequently, this project reinforced the fundamental value of relying on exact identifiers, establishing transparent thresholds, and clearly communicating analytical uncertainty. Throughout the process, intermediate validation metrics—such as row counts, null checks, candidate volumes, and score distributions—were meticulously tracked to help me evaluate whether the data pipeline was behaving sensibly.

Figures 1 and 2.Visual highlights



Monthly acquisition events show a long build-up, strong expansion after 2000, a late-2010s peak, and a sharp decline after 2020.



Counts were concentrated in a small number of offshore jurisdictions, led by Jersey and the British Virgin Islands.

Ultimately, from an investigative perspective, the most valuable outcome was not merely generating a list of named entities, but rather establishing a clearer decision rule regarding evidence quality. Exact company-number joins create an auditable trail that subsequent analysts can objectively verify; fuzzy name scores, conversely, do not. In a high-stakes sanctions context, where false positives carry severe compliance and reputational risks, this distinction is paramount. Therefore, the project successfully pivoted the primary question from ‘can these names be algorithmically manipulated to appear similar?’ to ‘what precise standard of evidence would reliably justify further investigative work?’. This fundamental shift in framing is, essentially, a highly accurate reflection of how I systematically approach complex analysis more broadly.

Limitations and Next Steps

If I were extending the project, I would prioritise broader entity-resolution features such as aliases, addresses, dates, and corporate relationship data rather than relying mainly on names. I would also refine the definition of ownership-change events and test the workflow on a larger sanctioned-entity set. Those steps would improve both precision and investigative

usefulness.

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